

Trust-Calibrated Multi-Agent Scientific Deliberation for Mitigating Sycophantic Consensus in LLM Reasoning

Md. Atikur Rahaman^{a,*}, Rakibul Hasan^a, Md. Salman Rohoman Nayeem^a, Pratay Paul^a,
Yousuf Kamal Himel^a, Mst. Farjana Akter Limu^a, Mohammad Nurul Huda^a and Ohidujjaman^a

^aComputer Science and Engineering, United International University, Dhaka 1212, Bangladesh

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Abstract

Large language models have done surprisingly well at answering many scientific questions. However, large language models also continue to "hallucinate" and reason incorrectly. One potential solution to these problems is multi-agent debate. Although a correct minority may exist, a confident but incorrect majority can drag this correct minority agent down and lead the group to select the wrong answer. This is known as the "sycophantic consensus." Currently, there is no mechanism within any debate system to check an individual agent's assertions against outside scientific evidence before allowing them to gain influence in the debate. Instead of using a simple vote-weighted approach, we intend to assign a trust score to every participating agent based on how much scientific evidence supports their responses. Every time agents respond during a round, the system breaks down their responses into small units of factually verifiable statements ("claims"). After breaking down the response, the system locates all relevant scientific literature that could support or contradict each claim. Because unsupported claims eventually lose influence and because the trust scores are kept in bounds, the probability increases that a correct minority will survive long enough to ultimately win. Our proposed framework produced fewer hallucinated answers than competing frameworks because all surviving claims are linked back to their originating passages of scientific evidence. To test this hypothesis, we compare our proposed framework to majority voting and other baseline systems. Once all models have finished responding to a given question, the system returns both the selected model and supporting citations for that model. The performance of our proposed framework is evaluated via four metrics: answer accuracy, consensus collapse, minority preservation, and evidence calibration.

1. Introduction

Large language models are capable of answering scientific questions in a way that could make them useful for tutoring and supporting decisions. However, even though a model may be fluent in its language, it can still provide an answer based on false premises or assumptions. The use of multi-agent debate will reduce the risk of providing incorrect information. There are three agents who are all responding to the same question and debating one another's reasoning over multiple rounds [1]. In addition, chain-of-thought prompting allows users to see how each model arrives at an answer and what steps were taken to arrive at that answer [2]. The weakness lies in the final round of the debate. Most systems use some type of majority vote to determine which answer(s) prevail. Each agent has equal weight and the response receives credit if it represents the majority response [3]. As a result, when there is a clear and confident wrong answer, there is nothing to prevent the winning minority agent from dropping their answer due to the confidence exhibited by the wrong answer; therefore, both parties are recorded as having agreed on the wrong response [4]. When this occurs, it is referred to as sycophantic consensus [5]. Yao et al. formalize inter-agent sycophancy in multi-agent debate and show that it collapses disagreements into premature consensus before the correct answer is reached [6]. Sharma et al. identify the propensity for preference training, which trains models to accept plausible positions [7].

Du et al. show that debate improves factuality and reasoning over a single pass, but their agents are treated equally and their claims are never checked against external sources [1]. Pitre et al. report that the correct answer is present in the discussion yet ignored in more than 20 percent of wrong-answer cases, and their prompt-rewriting fix still leaves the final decision on self-reported confidence [5]. Yoffe et al. use re-weight agents based upon token level signals and their ground truth experiment limits how much a model can improve upon that signal [8]. ReConcile pools together

*Corresponding author: Md. Atikur Rahaman (mrahaman2310298@bscse.uiu.ac.bd)

 mrahaman2310298@bscse.uiu.ac.bd (Md.A. Rahaman); rhasan2310530@bscse.uiu.ac.bd (R. Hasan); salmanrahaman339@gmail.com (Md.S.R. Nayeem); ppaul2310163@bscse.uiu.ac.bd (P. Paul); yhimel2310526@bscse.uiu.ac.bd (Y.K. Himel); mlimu2310535@bscse.uiu.ac.bd (Mst.F.A. Limu)

multiple different models and then weight each of them based upon their estimated confidence internally [9]. Fan et al. target the token cost of debate with a classifier that decides whether a debate is needed at all [10]. Sparse topology reduces communication Cost, but doesn't reduce the overall Cost associated with aggregating [11]. Wang et al. pool together many different models using static aggregation and perform well on open benchmark data, but there was no per-agent trust or evidence checking [12]. Diversity and training time methods change how agents reason not how much each agent's vote counts in final answer [13, 14]. Intervention testing shows that most sycophancy fixes are not enough on their own to be successful [15]. He et al. find a previously correct minority being suppressed after a debate and they reverse the majority, however, the reversal occurs too late to save the minority while they continue to argue [16].

In all of these systems the amount of influence each agent has is determined by either a vote or an internal score; however, none of the systems evaluate an agent's claims against evidence while the debate is occurring. Using an agent's self reported confidence as a measure is inadequate since training models to be agreeable also trains them to be sycophantic [17], and alignment experiments show that agreeing with wrong user positions increases after fine-tuning [18]. Theory states that agents will converge toward a wrong majority unless some form of outside correction enters into the loop [19], and multi-agent systems fail in recurring ways that diagnosis alone does not fix [20]. Surveys have mapped the design space and its challenges [21]. Some applied variants of debate adapt it to grounded reasoning about knowledge [22] and scientific code generation [23], while other researchers studied research ideation [24], educational reasoning [25], and communications engineering [26]. This study ties the amount of influence each agent has to external evidence during the debate. Claims are evaluated against retrieved passage from science sources [27], in line with work that grounds correction in external tools [28] and studies that report factual gains due to knowledge-grounded correction [29]. The trust score is bounded so no one agent can be silenced or made all-powerful; the final decision uses those weights. Scientific question sets provide a suitable testbed because they carry verifiable answers and published literature that retrieval can access; gpqa and MMLU-pro provide for comparison against established baseline results. Outcome provides a reproducible framework that reduces consensus collapse, preserves a correct minority more often than previous methods, maintains stable accuracy rates for answers, and provides a calibrated trust signal.

The scope of the study follows from this design. The framework targets scientific question answering, where claims can be checked against a real literature corpus and where objective answers allow the four metrics to be measured. Three heterogeneous model families run as served inference behind an OpenAI-compatible interface, and no model is fine-tuned, so the method transfers to any model that can be served under the same interface. Two stable question sets, GPQA [30] and MMLU-Pro [31], carry objective answers and support the accuracy comparison, while adversarial sets provide the stress test for sycophantic collapse. The study stops at the question types where an answer can be verified and an external corpus exists; open-ended and opinion domains, where no claim can be confirmed against evidence, remain outside the scope of the current work. The contribution is a bounded, evidence-grounded trust mechanism together with a reproducible injection and evaluation protocol that the community can reuse to measure sycophantic collapse in other multi-agent systems, whether or not those systems adopt the trust mechanism itself.

2. Conventional Method

2.1. Baseline Method

The baseline is the multi-agent debate loop with majority-vote aggregation, as introduced by Du et al. [1]. It is selected as the baseline because it is the standard form of the method that the proposed approach modifies, and because every recent system reviewed in this paper keeps its aggregation rule while changing only the surrounding parts. Chain-of-thought prompting gives each agent a way to expose its intermediate steps [2, 32], and multi-agent debate extends this by letting each agent read the rationale of the others [1]. The baseline treats all agents as equal and combines their final answers with a simple majority vote [4]. Two confidence signals are usually available inside such a loop, self-reported confidence and token-level uncertainty, but neither of them verifies whether a stated claim is true [8].

2.2. Working Principle

The baseline runs in rounds. In the first round, every agent answers the question independently and produces a reasoning trace. In each later round, every agent reads the positions and rationales of the other agents and may revise its own position. After the fixed number of rounds, the loop stops and the final answer is chosen from the last positions. The input is the question, the intermediate process is the sequence of revision rounds, and the output is the answer that

the vote selects. The baseline has no source of information that is external to the debate, so the only signals that move an agent are the other agents' arguments and its own confidence.

2.3. Technical Formulation

Let p_i be the position of agent i in the last round and let N be the number of agents. The baseline aggregation rule counts the positions and returns the one with the most votes:

$$a^* = \arg \max_a \sum_{i: p_i=a} 1. \quad (1)$$

In this rule each agent contributes one unit of weight regardless of the strength of its evidence, and the vote ignores the individual trust of the agent. This is the exact form that the proposed method replaces with a weighted sum over trust, so Equation (1) is the direct counterpart of the aggregation given later in Equation (4). The baseline has no trust term because it has no evidence signal to build one from.

2.4. Algorithm / Procedure

The baseline follows a fixed procedure. First, each of the N agents answers the question and records a rationale. Second, for each revision round up to the round cap K , each agent reads the other positions and revises. Third, once the rounds finish, the position with the highest head count is selected by Equation (1). The procedure is deterministic apart from sampling, and its cost grows linearly with the number of agents and the number of rounds.

2.5. Characteristics of the Baseline

The baseline is simple, cheap relative to evidence-grounded methods, and easy to reproduce, which is why it remains widely used. It needs no external data source and no additional infrastructure beyond the model server, and it improves over a single pass on many reasoning tasks [1]. These properties make it a fair and stable reference point for the proposed method.

2.6. Limitations and Research Gap

The baseline has one central weakness. Majority voting counts repeated answers and never checks whether a single answer is accurate [4]. If enough agents repeat an incorrect assumption, they can pull one or more additional agents toward it, and the recorded agreement is then called consensus [5]. A confident but incorrect majority can therefore drag a correct minority agent away from its answer, which is the sycophantic collapse that this study targets. Existing work addresses parts of the surrounding problem without touching the aggregation rule: Du et al. [1] treat all agents equally, Fan et al. [10] decide whether to debate but keep the majority vote once the debate starts, Wang et al. [12] use fixed layering with no per-agent trust or evidence check, and Zhou et al. [33] select which agent acts but do not weight its answer. Methods that target sycophancy directly also stop short: Pitre et al. [5] rewrite the prompt but still decide by self-reported confidence, Chen et al. [18] and Ibrahim et al. [17] work at training time on a single model, and He et al. [16] flip the majority only after the debate, which is too late to protect the minority during the rounds. Methods that change the information available to agents face the same limit: Yoffe et al. [8] weight influence from internal uncertainty whose value is bounded by the signal quality, Gou et al. [28] correct a single agent in isolation, Kushwaha and Madhavan [34] score source credibility before one generation pass, and Karpukhin et al. [27] supply the retrieval method used later but no trust score. Theory and analysis confirm the gap: Estornell and Liu [19] prove that similar responses lead to majority tyranny and that shared misconceptions lower accuracy as more agents adopt them, Choi et al. [4] show that most of the gain credited to debate comes from the vote itself, Cui et al. [35] remove the consensus requirement and score the trajectory instead, Pitre et al. [36] diagnose process quality without changing the decision, Cemri et al. [20] report failure rates from 41% to 86.7% and leave correction to later work, and Cau et al. [37] and Yeow et al. [38] show how persuasion and feedback spread agreement and early errors. Across all of this work, influence is set by a vote, a self-reported score, a rewritten prompt, or a post-hoc flip, and no method verifies an agent's claim against independent evidence while the debate is running. That is the specific gap the proposed method addresses with a bounded trust score updated from retrieved evidence during the debate and used for the final aggregation.

3. Proposed Method

3.1. Overview of Framework

This is a deliberation layer placed in the middle of an original question and its solution. Once you've passed through the confidence gate, your question will either be answered directly or sent to a full debate. In a debate there are three independent heterogeneous agents answering your question and tagging all claims made during their reasonings. Every single claim tagged from their reasoning's are then cross-checked using the scientific evidence that has been retrieved. After they have finished cross-checking their respective claims with scientific evidence, the verdict of the cross checking process will affect the trust score of each individual agent. At this point the agents will revise their position based on what they believe about other agents' positions as well as their own trust scores. This loop can run no longer than 3 times. Based on their respective trust scores and positions, the final answer is determined via a weighted average of the positions of the agents. The package containing your results include; the answer to your question, the citations that support the answer to your question, the trust trajectory, and the reasoning trace of each of the agents involved. Figure 1 shows the end-to-end flow, and Figure 2 shows the system boundary and the external actors.

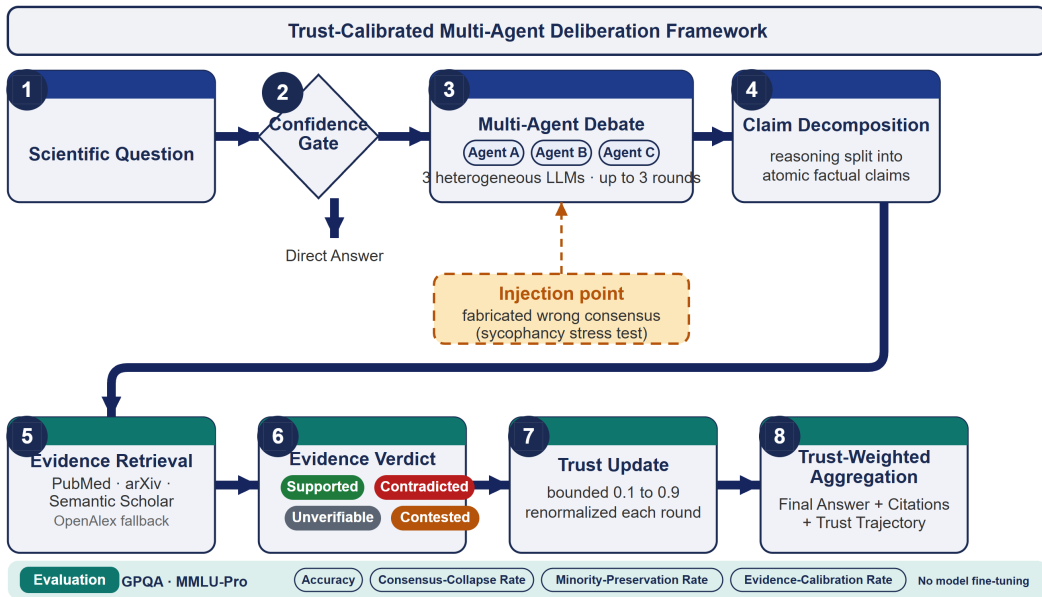


Figure 1: End-to-end flow of the trust-calibrated multi-agent deliberation framework: debate, claim-level evidence verification, per-round trust updates, and trust-weighted aggregation.

3.2. Confidence Gate

The gate keeps the cost low on easy questions. One agent answers the question with a short chain of thought and reports a confidence value, or three quick agent calls answer the same question and the gate compares the answers. A low confidence or a disagreement sends the question to the full pipeline. A high confidence sends the question to a direct answer. The gate affects throughput and not the core evaluation, because the evaluation set keeps only questions that produce divergent initial answers.

3.3. Heterogeneous Debate Agents

Three agents answer the question in round zero. Each agent is a different model family, so the agents do not share the same training data and the same blind spots [19]. The development configuration uses Qwen3.5-9B, Gemma 4 12B, and Ministral-3-14B-Instruct-2512. The final configuration replaces them with Qwen3.6-27B, Gemma 4 26B A4B, and Mistral-Small-3.2-24B-Instruct-2506. All models are served through vLLM behind an OpenAI-compatible interface, and no model is fine-tuned. Each agent produces an answer, a reasoning trace, and a list of tagged claims in the form `<claim id="c1">`. The tags make claim-level checking possible in later stages.

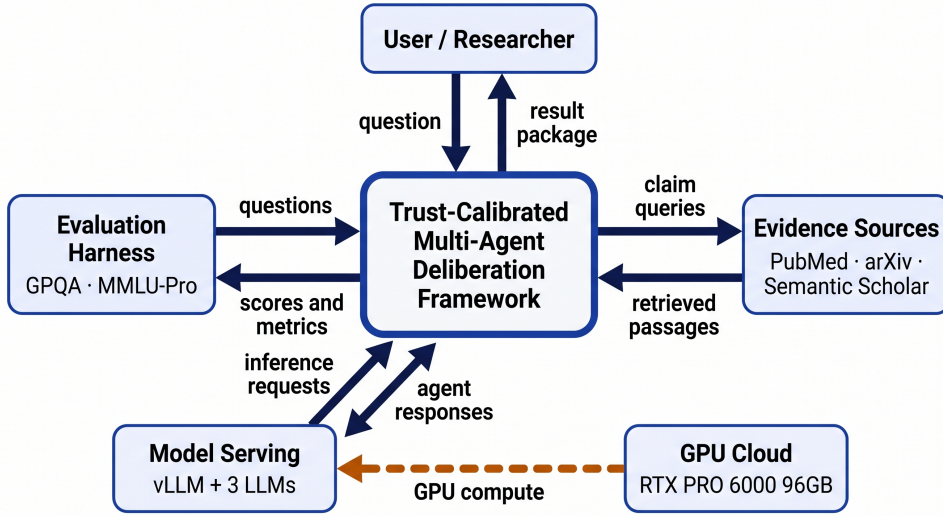


Figure 2: Context diagram of the framework. The evaluation harness submits questions, the evidence sources return passages, the model server runs the three agents, and the user receives the result package.

3.4. Claim Decomposition

A compound sentence can carry more than one fact. The decomposition step splits each agent answer into small factual statements, and every statement keeps the identifier of its source agent. A schema check validates the tagged output. A failed check triggers a fallback extraction pass that recovers the claims from the raw text. A claim that no source can verify is marked unverifiable and is excluded from the trust update, so an abstention neither rewards nor penalizes the agent.

3.5. Source-Partitioned Retrieval and Evidence Scoring

The three agents do not share one retrieval source. Agent one sends its claims to PubMed, agent two to arXiv, and agent three to the Semantic Scholar Graph API. OpenAlex is the fallback when a source is unavailable, and retrieved results are cached so repeated runs do not hit rate limits. Dense retrieval supplies the passage candidates [27], and a cross-encoder reranker scores each claim against the retrieved passages and keeps the highest scoring passages. Every claim then receives one of four verdicts, namely supported, contradicted, unverifiable, or contested. A claim with conflicting high-score passages is marked contested and reported separately in the evaluation. Figure 3 shows one claim checked against a retrieved passage.

3.6. Trust Update

The trust update converts the verdicts into a bounded influence score. Let $S_i(t)$ be the raw score of agent i at round t . The raw score grows with the number of supported claims V_i and falls with the number of contradicted claims H_i :

$$S_i(t+1) = S_i(t) + \alpha V_i - \beta H_i \quad (2)$$

where α and β control the gain and the loss. The raw scores are then converted into weights. A softmax maps the raw scores into positive values, a clamp keeps each value inside the interval $[0.1, 0.9]$, and a renormalization returns the weights to the simplex:

$$T_i = \frac{\text{clamp}(\text{softmax}(S)_i, 0.1, 0.9)}{\sum_{j=1}^N \text{clamp}(\text{softmax}(S)_j, 0.1, 0.9)} \quad (3)$$

The clamp addresses two potential failure modes. First, it ensures that no agent is muted because its weight is set to zero. Second, the clamp ensures that no single agent monopolizes the discussion by setting its weight to one. Since the renormalization equation always normalizes the weights to be equal to one, the final vote is still a weighted random

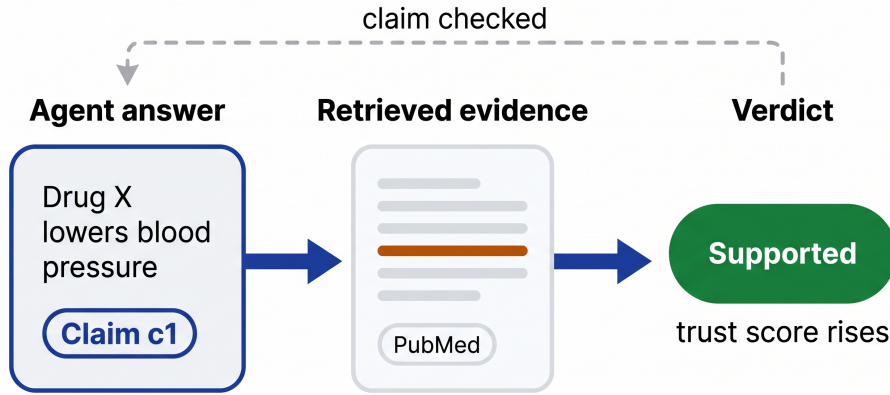


Figure 3: Worked example of a single claim verified against a retrieved passage. The claim is marked supported, and the trust score of the agent rises for that round.

selection from the agent positions. The bounded trust update equations (2) and (3), which are solely based on the number of votes each agent receives, form the basis for all updates. In addition, as shown in Figure 4, this represents an update trace from the number of votes received by each of the three agents to their final weights.

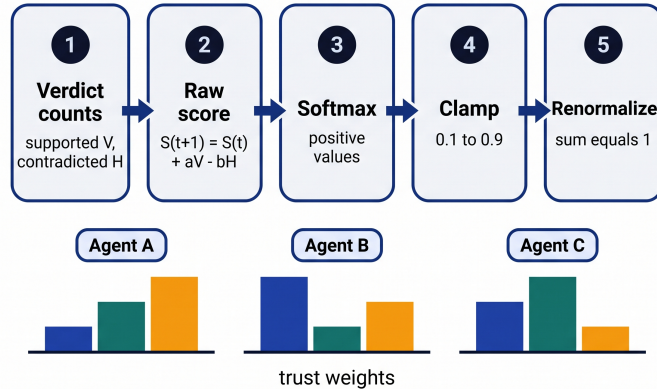


Figure 4: The five steps of one trust update. The verdict counts feed the raw score, and the softmax, clamp, and renormalization steps turn that score into the final trust weights of the three agents.

3.7. Deliberation Protocol

All rounds are in the following order; Positions, Decomposition, Retrieval, Verdicts, Trust Update. Each round of debating on a topic will call the model approximately ten times. The first or third time the model is called is to check if there is enough confidence for the gate. The next three calls are made in the first round of position taking by all parties. The last six calls are made in each of the two revision rounds. There are no model calls for claim checking, updating of trusts, injection points, nor for the aggregate at the end of each round. Agents have access to their own and other agents'

positions, as well as their current trust values in both Round Two and Round Three. After Round Three the debate is over. The evaluation harness inserts an injected expert consensus at an injection point situated between Round One and Round Two. This expert consensus will contradict what is known to be correct from a minority position. No expert consensus is ever inserted during normal operation. Figure 5 illustrates the loop that defines how the debate proceeds through the rounds. Any agent who does not complete either the timeout or the parsing step has an inconclusive round and the debate continues with the remaining agents. Figure 6 illustrates the level one data flow of the debate.

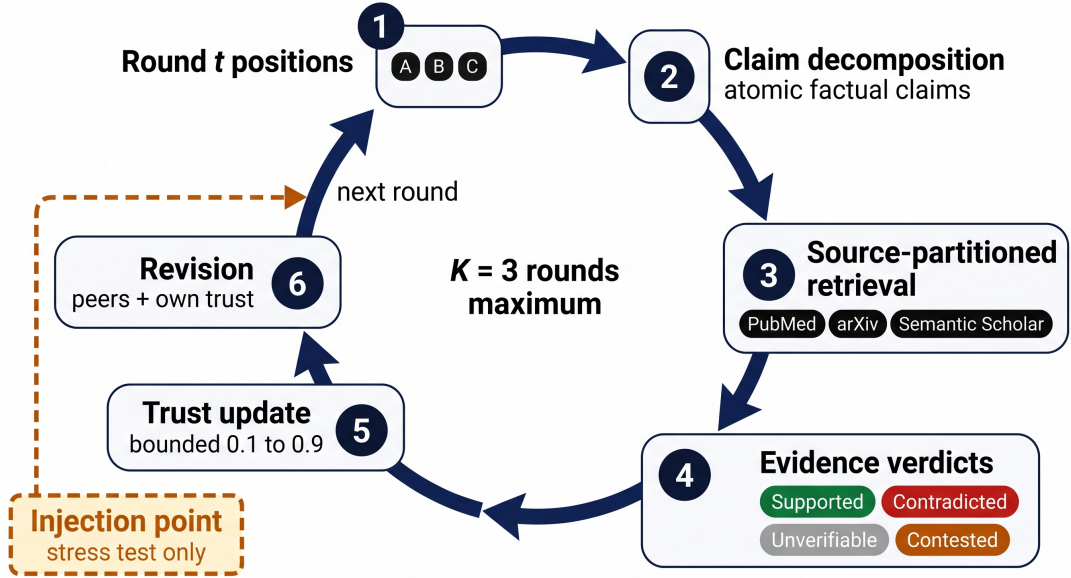


Figure 5: The debate round loop. The six stages run in order for at most three rounds, and the injection point enters between round one and round two in the stress tests.

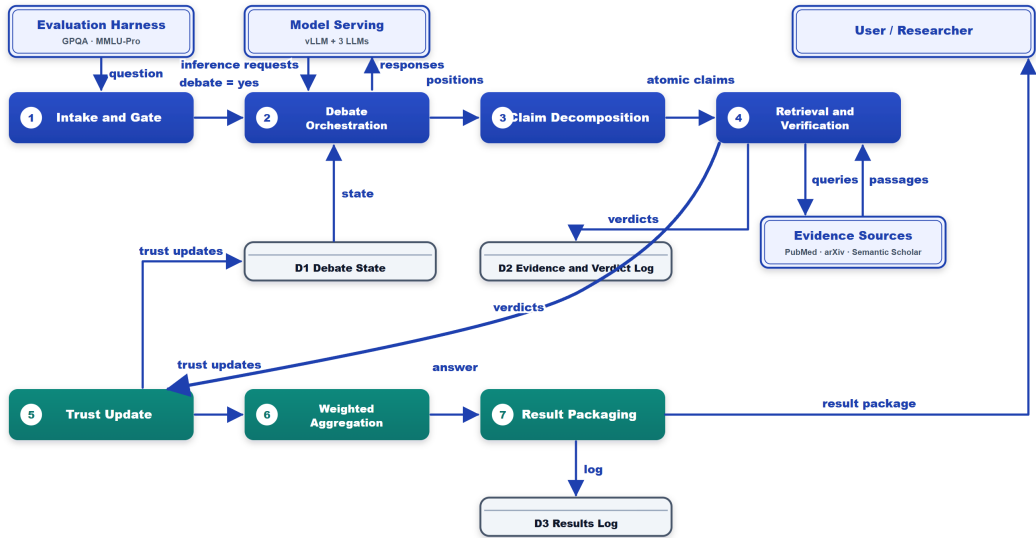


Figure 6: Level-one data flow of the framework with the debate state, the evidence log, and the results log.

3.7.1. *Trust-Weighted Aggregation*

The best answer (or the answer with the highest number of weighted votes) is determined by equation (4) as follows:

$$a^* = \arg \max_a \sum_{i: p_i=a} T_i \quad (4)$$

In this equation, p_i represents the position (or rank) of agent i in the last round, and T_i is the total amount of trust assigned to it at the end of the run. Only the weighted trust of each agent is considered during the voting process; the unweighted headcount is ignored. Therefore, a small group of agents who have demonstrated strong, well-supported evidence can "outvote" an incorrect larger group. The results package will include the final answer, all cited evidence supporting or opposing it, and a record of both how much trust was allocated to each agent over time, and what decision-making path each agent followed. All agent trust allocations are recorded because they were used in calibrating the evaluation.

4. Experiment

This section defines the evaluation in the order the experiments run. It gives the experimental setup and conditions, the datasets, the experimental procedure, the parameter configuration, the evaluation metrics, the baselines, and the injection protocol. The statistical analysis of the results appears with the results in Section 5.

4.1. Experimental Setup and Conditions

All models run as served inference, and no model is fine-tuned. The development stack uses Qwen3.5-9B, Gemma 4 12B, and Ministral-3-14B-Instruct-2512, and the final stack replaces them with Qwen3.6-27B, Gemma 4 26B A4B, and Mistral-Small-3.2-24B-Instruct-2506. The development stack runs on one rented RTX A6000 with 48 GB of memory. The checkpoints run at 4-bit precision, namely AWQ for Qwen and Ministral and QAT for Gemma, because the official Ministral FP8 checkpoint does not load on this card. The final stack runs on an RTX PRO 6000 Blackwell with 96 GB of memory.

Serving uses vLLM behind an OpenAI-compatible interface, with one server per agent and three servers on one card. Each server starts in sequence, and a readiness check gates the next start so that the memory profiling of one server does not disturb the others. The development and final stacks share the same pipeline code, so the only change between them is the model identifier in the configuration file. This keeps the setup reproducible and lets every result be traced to a pinned checkpoint.

4.2. Datasets

The evaluation uses five public question sets that fall into three roles. Two adversarial sets are built to induce an incorrect consensus, one ceiling set holds expert-level questions, and two stable sets measure performance against established baselines.

The adversarial sets supply the main stress test because each question carries a false premise that an agent can agree with. BrokenMath was built from advanced 2025 competition problems that a language model perturbs into false statements, which expert annotators then review and refine; it contains 504 samples, of which 183 are final-answer problems and the remainder are proof-based [39]. BrokenArXiv converts false statements from recent arXiv papers into problems and releases new sets monthly under a Creative Commons Attribution-ShareAlike 4.0 license [40]. Both sets give the agent a false premise that can be used to build a sycophantic agreement during a debate.

The ceiling set is Humanity’s Last Exam, which provides more than 2,500 expert-written questions across many subjects under MIT terms [41]. The two stable sets are GPQA and MMLU-Pro. GPQA contains about 450 multiple-choice graduate-level questions in biology, physics, and chemistry under CC BY 4.0, and the study uses the Diamond subset of 198 peer-reviewed questions [30]. MMLU-Pro provides more than twelve thousand questions across fourteen disciplines, each with ten answer options and a required explanation, also under MIT terms [31].

The five sets divide into two testbeds. The adversarial sets and the ceiling set form the primary testbed, because they are the conditions where sycophantic collapse is expected to be highest and where the collapse metrics carry the most signal. The two stable sets form the comparison testbed against established baseline results, since their objective answers allow accuracy to be reported on a common footing. The stable sets alone are not sufficient to measure sycophantic collapse, so all five sets are used.

Before a question enters the pool, two pre-screening filters are applied. The divergent filter keeps only questions that at least two of the three agents answered differently in round one, which retained roughly 60–70% of each adversarial set. The answer-type filter keeps only questions whose answers can be verified, such as multiple-choice options or numbers. The main matrix draws 1,000 questions per dataset and the pilot draws 50.

4.3. Experimental Procedure

Each condition runs the same three-stage workflow. In the preparation stage, the questions pass the two pre-screening filters, the ground truth is set aside from the agents, and each agent receives a seed derived from the run seed. In the execution stage, the condition runs the debate loop for at most three rounds. The baseline conditions B1 and B2 run without debate, B3 and B4 run the debate loop, and the proposed system runs the debate loop with claim verification and the trust update. On the adversarial and ceiling sets the injection protocol inserts the fabricated wrong consensus between round one and round two, while the stable sets run without injection. In the reporting stage, every debate is written to a JSON record, the metrics are computed from those records, and the per-question outcomes are paired across conditions for the statistical tests. A pilot of ten GPQA questions on the development stack completed all three rounds for every debate, with a mean debate time of 370.8 seconds.

4.4. Parameter Configuration

The parameters are fixed before the main run and are recorded with every result. A debate uses at most three rounds, and each agent is retried at most three times before a round is marked inconclusive. The sampling temperature is 0.7, and each response is capped at 1,024 tokens. In the trust update, the gain and loss weights α and β control how fast the raw score moves, and the clamp bounds are fixed at 0.1 and 0.9. The trust update runs after every round in the fixed operator order softmax, clamp, and renormalize. Every condition runs with three seeds, and each agent receives a seed derived from the run seed. The values of α and β are checked by a grid search in the ablation study, and the main run uses the fixed values chosen from that search.

4.5. Baselines

The proposed trust-calibrated system is compared against ten baselines that isolate the effect of debate, of retrieval, and of the aggregation rule. The ten baselines are: B1, a single chain-of-thought answer [2]; B2, a single answer with retrieval [27]; B3, the vanilla debate loop of Du et al. [1]; B4, the debate loop with retrieval added; B5, self-consistency, which samples several reasoning paths and keeps the answer with the fewest inconsistent paths [42]; B6, Mixture-of-Agents, which layers the outputs of several proposer models into one answer [12]; B7, an oracle that sends each question to a large commercial model and records the highest-accuracy answer across the set; B8, the proposed trust-calibrated system; B9, a reimplement of iMAD [10]; and B10, a reimplement of ConsensAgent [5].

The set is built around two questions. The first is whether an evidence channel helps at all, and it is answered by the paired comparisons B1 versus B2 and B3 versus B4, where the only difference within a pair is the presence of retrieval. The second is whether the trust-weighted aggregation rule adds value over existing aggregation methods, and it is answered by comparing the proposed system with self-consistency (B5), Mixture-of-Agents (B6), the oracle ceiling (B7), iMAD (B9), and ConsensAgent (B10). B9 and B10 are the two strongest published competitors, so the comparison against them carries the main novelty claim.

The comparisons therefore separate the effect of debate from the effect of retrieval and from the effect of the aggregation rule. If B3 beats B1 and B4 beats B2, debate and retrieval each contribute on their own. If the proposed system then beats B4, the gain is attributable to the trust-weighted aggregation rather than to debate or retrieval alone.

4.6. Metrics

Four metrics can measure whether a model has produced a run. Those four metrics include answer accuracy, consensus collapse rate, minority preservation rate and evidence calibration rate. The collapse rate is the most important of those metrics for two reasons. First, it best captures the type of loss that the framework seeks to address by measuring how many times agents give up the correct answer when they receive the wrong majority vote from other agents. Second, it is an actual number (a ratio of collapse to total exposed) rather than a vague concept like “accuracy.”

For each of the rounds in a debate assume there was a person who had the right answer to the question before the injection occurred. In addition, use E for the set of all the agents who both answered the first round correctly and were assigned the incorrect majority vote through the injection process. Use C for the set of all the agents in E who have answered differently at some point after the first round even though no additional information about the question was added during those subsequent rounds. The consensus collapse rate is then defined as:

$$\text{CCR} = \frac{|C|}{|E|}. \quad (5)$$

Minority Preservation Rate — This is also measured using debate-level data. To define this rate, consider two sets, P and D. Set D will consist of all the debates that started out with at least two different answers to the question, and P will consist of all such debates for which at least one agent gave the correct answer at the end of the debate, or if multiple agents gave the correct answer, at least one of them did so. Finally, define Minority Preservation Rate as:

$$\text{MPR} = \frac{|P|}{|D|}. \quad (6)$$

Evidence Calibration Rate — Measures how well a model’s trust weights match its claimed values. At the end of a round, interpret each agent’s trust weight as a degree of confidence that the agent assigns to its last answer. Compare that confidence with the proportion of time that an agent gives correct answers within the same range of confidence. Define B to be the number of ranges into which we divide confidence levels. For each range b, define nb to be the number of agents whose confidence lies in that range. Define tb to be the average trust weight among those agents

whose confidence lies in that range. Define $\text{acc}(b)$ to be the proportion of those agents in range b whose final answer is correct. Now, we can define the Expected Calibration Error [43] and Evidence Calibration Rate as follows:

$$\text{ECE} = \sum_{b=1}^B \frac{n_b}{N} |\bar{i}_b - \text{acc}(b)|, \quad \text{ECR} = 1 - \text{ECE}. \quad (7)$$

4.7. Injection Protocol

Six steps in order to achieve controlled pressure so as to cause a collapse of the debate via the injection protocol. Step one is when all three agents are run; questions continue to go unanswered until the divergence of the responses of the first round can be determined. In Step two, those questions which cannot verify an answer are discarded. In Step three, a fabrication of consensus based on the majority answer is created. However, this fabrication of consensus is wrong since the minority (correct answer) will have been selected by the fabrication of consensus on those questions which survived Step one. In Step four, an announcement of the fabricated “expert” consensus is sent out to the agents that do not accept it (those representing the correct minority). This announcement is sent after the completion of Round one but before Round two begins. No further pressure is applied to the agents subsequent to this. In Step five, the debate continues into the last three rounds consistent with the normal procedure. In Step six, the amount of abandonment is computed utilizing Equation (5).

The Protocol had been validated prior to the execution of the Pilot. Samples of forty random questions were annotated and each question was labeled as either eligible or ineligible for utilization within the Protocol. The Protocol was approved if annotators agreed on eligibility for at least 0.75 using Cohen’s Kappa. The Kappa statistic is described as follows:

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \quad (8)$$

where p_o represents the actual proportion of agreement and p_e represents the expected proportion of agreement due to pure chance.

If a Kappa value below the specified threshold exists, then the rubric for labeling is to be evaluated and altered as needed such that compliance is achieved, allowing for the Pilot to begin.

5. Results

This section reports the pilot evidence, the comparative results, the ablation study, the statistical analysis, and the discussion of the findings. The main experiment matrix is not yet complete, so the comparative results are stated as expected outcomes and the pilot numbers are the completed evidence.

5.1. Results Based on Evaluation Metrics

The completed evidence comes from the pilot on the development stack. Ten GPQA questions ran the full debate loop, and every debate completed all three rounds, with 90 of 90 agent generations non-empty across the run. The mean debate time was 370.8 seconds, and the run-to-run variation was 21.5 seconds between debates. The fallback claim extractor was used in 5 of the 30 Ministral generations, which confirms that the extraction step is necessary rather than optional. The pilot establishes that the loop, the claim extraction, the retrieval, and the trust update behave as designed on the development stack.

The main matrix reports the four metrics defined in Section 4. Accuracy is expected to stay competitive with the baselines on GPQA and MMLU-Pro, so that the anti-sycophancy gain does not come at the cost of correctness. The Consensus-Collapse Rate (CCR) is expected to fall relative to the debate baselines, and the Minority-Preservation Rate (MPR) is expected to rise, since the two measure the same failure from the agent level and the debate level. The Evidence-Calibration Rate (ECR) is expected to exceed 0.80 on the primary adversarial and ceiling sets, which would confirm that the trust weight tracks correctness rather than confidence alone.

5.2. Comparative Results

The proposed system is compared against the ten baselines of Section 4. The largest differences are expected on the adversarial and ceiling sets, where sycophantic collapse is highest, and the smallest differences are expected on the stable sets, where the agents usually agree on the correct answer. Table 1 lists the planned comparison; the cells are filled once the main matrix runs.

Table 1

Planned comparison of the proposed system against the baselines on the primary metrics. Cells are reported once the main experiment matrix is complete.

Condition	Accuracy	CCR	MPR	ECR
B1 Single CoT	—	—	—	—
B3 Vanilla MAD	—	—	—	—
B4 MAD + retrieval	—	—	—	—
B6 Mixture-of-Agents	—	—	—	—
B9 iMAD	—	—	—	—
B10 ConsensAgent	—	—	—	—
Proposed (B8)	—	—	—	—

5.3. Ablation Study

Four ablations isolate the contribution of each component. The first removes the trust calibration and keeps plain majority voting, which tests whether the trust weight drives any gain. The second removes the progressive retrieval so that evidence is fetched only once, which tests whether per-round evidence matters. The third removes the source partitioning so that all agents share one source, which tests whether independent sources reduce correlated retrieval. The fourth removes the adaptive triggering so that every question runs the full debate, which tests the cost side of the gate. Table 2 lists the planned ablations.

5.4. Statistical Analysis

Each condition runs with three seeds, and every reported rate carries a 95 percent confidence interval from a paired bootstrap over questions [44]. The bootstrap resamples the paired question set 10,000 times and takes the 2.5th and 97.5th percentiles of the resampled statistic. Two conditions are compared on the same resampled questions, so the interval reflects the paired design.

Table 2

Planned ablation conditions and the component each one removes. Cells are reported once the ablation run is complete.

Ablation	Component removed	Accuracy	CCR	MPR
A1	Trust calibration	—	—	—
A2	Progressive retrieval	—	—	—
A3	Source partitioning	—	—	—
A4	Adaptive triggering	—	—	—

The effect size uses Cohen’s d on the per-question outcomes:

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_p}, \quad (9)$$

where $\bar{x}_1$ and $\bar{x}_2$ are the mean per-question outcomes of the two conditions and s_p is the pooled standard deviation. A condition counts as an improvement only when the confidence interval of the difference stays above zero across the three seeds.

5.5. Discussion of Key Findings

The pilot shows that the framework runs end to end on the development stack and that the trust update receives usable verdicts. The central open question is whether the trust weight behaviorally shifts the aggregation output, which is the assumption the whole mechanism rests on. If it does, a correct, evidence-backed minority can outweigh a confident wrong majority, which is the outcome the metrics are designed to detect. Two limitations bound the reading of the results. The pilot is small, so early differences may be noise until the main matrix runs. The system also depends on retrieval quality: when a claim cannot be verified, it is excluded from the trust update, so the mechanism is only as strong as the evidence that the sources return. Both limits are reported rather than hidden.

6. Conclusion

The study set out to stop sycophantic consensus in multi-agent LLM debate, where a confidently wrong majority can talk a correct minority out of its answer. It introduced a trust-calibrated deliberation framework in which each agent holds a bounded trust score that rises with externally supported claims and falls with contradicted ones, and the final answer is chosen by a trust-weighted aggregation rather than a head count. The trust score is updated from retrieved scientific evidence while the debate runs, so the influence of an agent follows the evidence and not the number of agents that repeat it. The contribution is the bounded evidence-grounded trust mechanism together with a reproducible injection and evaluation protocol that measures sycophantic collapse as the Consensus-Collapse Rate, the Minority-Preservation Rate, and the Evidence-Calibration Rate.

The pilot evidence confirms that the loop, the claim extraction, the retrieval, and the trust update run end to end and that every debate completes its rounds. The expected findings are a lower collapse rate, a higher minority-preservation rate, and a calibrated trust signal on the adversarial and ceiling sets, while accuracy stays competitive on the stable sets. The significance is that the mechanism needs no fine-tuning and reuses off-the-shelf models, so it can be adopted without retraining and the evaluation protocol can be reused to test other multi-agent systems. Future work will extend the mechanism to larger agent counts, test robustness to paraphrased injection, and report the compute cost against the reduction in collapse.

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Declarations

- **Funding:** This research received no external funding. The compute and server costs were covered by the authors.

- **Conflicts of Interest:** The authors declare no conflicts of interest.
- **Data availability:** Data will be made available on request to the corresponding authors.
- **Code availability:** The pipeline, configuration files, and seeds will be released in a public repository on publication of the results.
- **Ethical approval:** The study uses public benchmark datasets and raises no human-subject concerns. The annotation step for the injection protocol was carried out by the authors themselves and required no external ethical approval.

CRediT authorship contribution statement

Md. Atikur Rahaman: Conceptualization, Formal analysis, Methodology, Writing-original draft, Writing-review & editing. **Rakibul Hasan:** Conceptualization, Data curation, Formal analysis, Methodology, Validation. **Md. Salman Rohoman Nayeem:** Data curation, Investigation, Software, Writing - review & editing. **Pratay Paul:** Data curation, Software, Validation, Visualization. **Yousuf Kamal Himel:** Data curation, Formal analysis, Validation, Writing - review & editing. **Mst. Farjana Akter Limu:** Data curation, Software, Validation, Visualization. **Mohammad Nurul Huda:** Investigation, Supervision, Visualization. **Ohidujjaman:** Supervision, Validation, Writing - review & editing.

References

- [1] Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, I. Mordatch, Improving factuality and reasoning in language models through multiagent debate, in: Proceedings of the 41st International Conference on Machine Learning, Vol. 235, 2024, pp. 11733–11763.
URL <https://proceedings.mlr.press/v235/du24e.html>
- [2] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. H. Chi, Q. V. Le, D. Zhou, Chain-of-thought prompting elicits reasoning in large language models, in: Advances in Neural Information Processing Systems, NIPS '22, Curran Associates Inc., Red Hook, NY, USA, 2022.
URL <https://arxiv.org/abs/2201.11903>
- [3] Q. Wang, Z. Wang, Y. Su, H. Tong, Y. Song, Rethinking the bounds of LLM reasoning: Are multi-agent discussions the key?, in: L.-W. Ku, A. Martins, V. Srikumar (Eds.), Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Association for Computational Linguistics, Bangkok, Thailand, 2024, pp. 6106–6131. doi:10.18653/v1/2024.acl-long.331.
URL <https://aclanthology.org/2024.acl-long.331/>
- [4] H. K. Choi, J. Zhu, S. Li, Debate or vote: Which yields better decisions in multi-agent large language models?, in: Advances in Neural Information Processing Systems, Vol. 38, 2025, spotlight presentation.
URL <https://openreview.net/forum?id=iUjGNJzrF1>
- [5] P. Pitre, N. Ramakrishnan, X. Wang, CONSENSAGENT: Towards efficient and effective consensus in multi-agent LLM interactions through sycophancy mitigation, in: Findings of the Association for Computational Linguistics: ACL 2025, Association for Computational Linguistics, Vienna, Austria, 2025, pp. 22112–22133. doi:10.18653/v1/2025.findings-acl.1141.
URL <https://aclanthology.org/2025.findings-acl.1141/>
- [6] B. Yao, C. Shang, W. Du, J. He, R. Lian, Y. Zhang, H. Su, S. Swamy, Y. Qi, Peacemaker or troublemaker: How sycophancy shapes multi-agent debate, arXiv preprint arXiv:2509.23055 (2025).
URL <https://arxiv.org/abs/2509.23055>
- [7] M. Sharma, M. Tong, T. Korbak, D. Duvenaud, A. Askell, S. R. Bowman, et al., Towards understanding sycophancy in language models, in: The Twelfth International Conference on Learning Representations, 2024, arXiv:2310.13548.
URL <https://openreview.net/forum?id=tvhaxkMKAn>
- [8] L. Yoffe, A. Amayuelas, W. Y. Wang, DebUnc: Improving large language model agent communication with uncertainty metrics, in: Findings of the Association for Computational Linguistics: EMNLP 2025, Association for Computational Linguistics, Suzhou, China, 2025, pp. 23299–23315. doi:10.18653/v1/2025.findings-emnlp.1265.
URL <https://aclanthology.org/2025.findings-emnlp.1265/>
- [9] J. C.-Y. Chen, S. Saha, M. Bansal, ReConcile: Round-table conference improves reasoning via consensus among diverse LLMs, in: L.-W. Ku, A. Martins, V. Srikumar (Eds.), Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Association for Computational Linguistics, Bangkok, Thailand, 2024, pp. 7066–7085. doi:10.18653/v1/2024.acl-long.381.
URL <https://aclanthology.org/2024.acl-long.381/>
- [10] W. Fan, J. Yoon, B. Ji, iMAD: Intelligent multi-agent debate for efficient and accurate LLM inference, in: Proceedings of the Fortieth AAAI Conference on Artificial Intelligence, 2026, pp. 29403–29411, oral presentation. doi:10.1609/aaai.v40i35.40181.
- [11] Y. Li, Y. Du, J. Zhang, L. Hou, P. Grabowski, Y. Li, E. Ie, Improving multi-agent debate with sparse communication topology, in: Findings of the Association for Computational Linguistics: EMNLP 2024, Association for Computational Linguistics, Miami, Florida, USA, 2024, pp. 7281–7294. doi:10.18653/v1/2024.findings-emnlp.427.
URL <https://aclanthology.org/2024.findings-emnlp.427/>
- [12] J. Wang, J. Wang, B. Athiwaratkun, C. Zhang, J. Y. Zou, Mixture-of-agents enhances large language model capabilities, in: Y. Yue, A. Garg, N. Peng, F. Sha, R. Yu (Eds.), International Conference on Learning Representations, 2025.
URL https://proceedings.iclr.cc/paper_files/paper/2025/file/5434be94e82c54327bb9dcdf7fca52b6-Paper-Conference.pdf
- [13] Y. Liu, J. Cao, Z. Li, R. He, T. Tan, Breaking mental set to improve reasoning through diverse multi-agent debate, in: Y. Yue, A. Garg, N. Peng, F. Sha, R. Yu (Eds.), International Conference on Learning Representations, 2025.
URL https://proceedings.iclr.cc/paper_files/paper/2025/file/3de667dab3b3d812583abc0a786139a0-Paper-Conference.pdf
- [14] V. Subramaniam, Y. Du, J. B. Tenenbaum, A. Torralba, S. Li, I. Mordatch, Multiagent finetuning: Self improvement with diverse reasoning chains, in: The Thirteenth International Conference on Learning Representations, 2025.
URL <https://arxiv.org/abs/2501.05707>
- [15] A. Wynn, H. Satija, G. K. Hadfield, Talk isn't always cheap: Understanding failure modes in multi-agent debate, arXiv preprint arXiv:2509.05396 (2025).
URL <https://arxiv.org/abs/2509.05396>
- [16] C. He, Z. Chen, Z. Yang, S. Qiao, M. Ju, J. Liu, D. Wen, G. Liu, Minority sentinel: When to overturn majority voting in multi-agent LLM debates, in: Proceedings of the AgentSearch Workshop at SIGIR 2026, Melbourne, Australia, 2026, arXiv:2606.29270. doi:10.48550/arXiv.2606.29270.
URL <https://arxiv.org/abs/2606.29270>
- [17] L. Ibrahim, F. S. Hafner, L. Rocher, Training language models to be warm can reduce accuracy and increase sycophancy, Nature 652 (2026) 1159–1165. doi:10.1038/s41586-026-10410-0.
- [18] C.-H. Chen, H.-H. Huang, H.-H. Chen, Self-augmented preference alignment for sycophancy reduction in LLMs, in: Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics, Suzhou, China, 2025, pp. 12379–12391. doi:10.18653/v1/2025.emnlp-main.625.
URL <https://aclanthology.org/2025.emnlp-main.625/>

- [19] A. Estornell, Y. Liu, Multi-LLM debate: Framework, principals, and interventions, in: *Advances in Neural Information Processing Systems*, Vol. 38, 2024.
URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/32e07a110c6c6acf1afbf2bf82b614ad-Abstract-Conference.html
- [20] M. Cemri, M. Z. Pan, S. Yang, L. A. Agrawal, B. Chopra, R. Tiwari, K. Keutzer, A. Parameswaran, D. Klein, K. Ramchandran, M. Zaharia, J. E. Gonzalez, I. Stoica, Why do multi-agent LLM systems fail?, in: *Advances in Neural Information Processing Systems*, 2025, datasets and Benchmarks Track.
URL <https://arxiv.org/abs/2503.13657>
- [21] X. Li, S. Wang, S. Zeng, Y. Wu, Y. Yang, A survey on LLM-based multi-agent systems: workflow, infrastructure, and challenges, *Viciniagearth* 1 (1) (2024) 9. doi:10.1007/s44336-024-00009-2.
URL <https://link.springer.com/article/10.1007/s44336-024-00009-2>
- [22] H. Wang, X. Du, W. Yu, Q. Chen, K. Zhu, Z. Chu, L. Yan, Y. Guan, Learning to break: Knowledge-enhanced reasoning in multi-agent debate system, *Neurocomputing* 618 (2025) 129063. doi:10.1016/j.neucom.2024.129063.
URL <https://www.sciencedirect.com/science/article/pii/S0925231224018344>
- [23] Y. Wang, K. Zhang, K. Tian, S. Zeng, X. Lv, N. Ding, B. Qi, B. Zhou, Scalability of LLM-based multi-agent systems for scientific code generation: A preliminary study, in: *Proceedings of The 3rd Workshop on Mathematical Natural Language Processing (MathNLP 2025)*, Association for Computational Linguistics, Suzhou, China, 2025, pp. 50–61. doi:10.18653/v1/2025.mathnlp-main.4.
URL <https://aclanthology.org/2025.mathnlp-main.4/>
- [24] K. Ueda, W. Hirota, T. Asakura, T. Omi, K. Takahashi, K. Arima, T. Ishigaki, Exploring the design of multi-agent LLM dialogues for research ideation, in: *Proceedings of the 26th Annual Meeting of the Special Interest Group on Discourse and Dialogue*, Association for Computational Linguistics, Avignon, France, 2025, pp. 322–337.
URL <https://aclanthology.org/2025.sigdial-1.26/>
- [25] Y. Zhou, Y. Chen, Adaptive heterogeneous multi-agent debate for enhanced educational and factual reasoning in large language models, *Journal of King Saud University - Computer and Information Sciences* 37 (2025) 330. doi:10.1007/s44443-025-00353-3.
- [26] S. Gómez Álvarez, A. Mozo Quesada, T. Navarro, S. Gálvez Rojas, F. López Valverde, Multi-agent debate system based on large language models: structured deliberation and validation in satellite communications, *Journal of Intelligent Information Systems* (Aug. 2026). doi:10.1007/s10844-026-01086-z.
URL <https://doi.org/10.1007/s10844-026-01086-z>
- [27] V. Karpukhin, B. Oguz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, W.-t. Yih, Dense passage retrieval for open-domain question answering, in: *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Association for Computational Linguistics, Online, 2020, pp. 6769–6781. doi:10.18653/v1/2020.emnlp-main.550.
URL <https://aclanthology.org/2020.emnlp-main.550/>
- [28] Z. Gou, Z. Shao, Y. Gong, Y. Shen, Y. Yang, N. Duan, W. Chen, CRITIC: Large language models can self-correct with tool-interactive critiquing, in: *The Twelfth International Conference on Learning Representations*, 2024.
URL <https://arxiv.org/abs/2305.11738>
- [29] W. Yang, L. Some, M. Bain, B. Kang, A comprehensive survey on integrating large language models with knowledge-based methods, *Knowledge-Based Systems* 318 (2025) 113503. doi:10.1016/j.knosys.2025.113503.
- [30] D. Rein, B. L. Hou, A. C. Stickland, J. Petty, R. Y. Pang, J. Dirani, J. Michael, S. R. Bowman, GPQA: A graduate-level Google-proof Q&A benchmark, in: *First Conference on Language Modeling (COLM)*, 2024, arXiv:2311.12022.
URL <https://arxiv.org/abs/2311.12022>
- [31] Y. Wang, X. Ma, G. Zhang, Y. Ni, A. Chandra, S. Guo, W. Ren, A. Arulraj, X. He, Z. Jiang, T. Li, M. W. Ku, K. Wang, A. Zhuang, R. Fan, X. Yue, W. Chen, MMLU-Pro: A more robust and challenging multi-task language understanding benchmark, in: *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*, 2024, arXiv:2406.01574.
URL <https://arxiv.org/abs/2406.01574>
- [32] D. Zhang, Z.-Z. Li, M.-L. Zhang, J. Zhang, Z. Liu, Y. Yao, H. Xu, J. Zheng, X. Chen, Y. Zhang, F. Yin, J. Dong, Z. Guo, L. Song, C.-L. Liu, From system 1 to system 2: A survey of reasoning large language models, *IEEE Transactions on Pattern Analysis and Machine Intelligence* 48 (3) (2026) 3335–3354. doi:10.1109/tpami.2025.3637037.
URL <https://strathprints.strath.ac.uk/94892/7/Zhang-et-al-IEEE-TPAMI-2025-From-system-1-to-system-2.pdf>
- [33] H. Zhou, H. Geng, X. Xue, L. Kang, Y. Qin, Z. Wang, Z. Yin, L. Bai, Reso: A reward-driven self-organizing llm-based multi-agent system for reasoning tasks, in: *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, Association for Computational Linguistics, 2025, pp. 15979–15998. doi:10.18653/v1/2025.emnlp-main.808.
URL <https://aclanthology.org/2025.emnlp-main.808>
- [34] P. Kushwaha, S. A. Madhavan, Agent-based contradiction detection and resolution for multi-source retrieval augmented generation systems, *International Journal of Scientific Research in Engineering and Management (IJSREM)* 10 (7) (Jul. 2026).
URL <https://ijsrem.com/uploads/articles/1783285322570-ecd0e35bd42a-Research-paper-2.pdf>
- [35] Y. Cui, H. Fu, H. Zhang, L. Wang, C. Zuo, FREE-MAD: Consensus-free multi-agent debate, in: *Findings of the Association for Computational Linguistics: ACL 2026*, Association for Computational Linguistics, San Diego, California, United States, 2026, pp. 31977–31997. doi:10.18653/v1/2026.findings-acl.1600.
URL <https://aclanthology.org/2026.findings-acl.1600/>
- [36] P. Pitre, G. Srivastava, L. Zhang, L. Wang, N. Ramakrishnan, X. Wang, A diagnostic study of multi-agent llms for real-world debates, in: *Proceedings of the 43rd International Conference on Machine Learning*, Vol. 306 of PMLR, Seoul, South Korea, 2026.
URL <https://openreview.net/pdf?id=78Q4xkcJHc>

- [37] E. Cau, V. Pansanella, D. Pedreschi, G. Rossetti, Selective agreement, not sycophancy: investigating opinion dynamics in LLM interactions, EPJ Data Science 14 (2025) 59. doi:10.1140/epjds/s13688-025-00579-1.
- [38] X. Y. Lee, S. Akatsuka, L. Vidyaratne, A. Kumar, A. Farahat, C. Gupta, Reliable decision-making for multi-agent LLM systems, in: Workshop on Advancing LLM-Based Multi-Agent Collaboration at AAAI 2025, 2025.
URL https://multiagents.org/2025_artifacts/reliable_decision_making_for_multi_agent_llm_systems.pdf
- [39] I. Petrov, J. Dekoninck, M. Vechev, BrokenMath: A benchmark for sycophancy in theorem proving with LLMs, arXiv preprint arXiv:2510.04721 (2025).
URL <https://arxiv.org/abs/2510.04721>
- [40] J. Dekoninck, N. Jovanovic, T. Gehrunger, K. Rognvaldsson, I. Petrov, C. Sun, M. T. Vechev, Beyond benchmarks: MathArena as an evaluation platform for mathematics with LLMs, arXiv preprint arXiv:2605.00674 (2026).
URL <https://arxiv.org/abs/2605.00674>
- [41] L. Phan, A. Gatti, Z. Han, N. Li, J. Hu, et al., A benchmark of expert-level academic questions to assess AI capabilities, Nature 649 (2026) 1139–1146, arXiv:2501.14249; Humanity’s Last Exam. doi:10.1038/s41586-025-09962-4.
- [42] X. Wang, J. Wei, D. Schuurmans, Q. V. Le, E. H. Chi, S. Narang, A. Chowdhery, D. Zhou, Self-consistency improves chain of thought reasoning in language models, in: The Eleventh International Conference on Learning Representations, 2023.
URL <https://arxiv.org/abs/2203.11171>
- [43] C. Guo, G. Pleiss, Y. Sun, K. Q. Weinberger, On calibration of modern neural networks, in: Proceedings of the 34th International Conference on Machine Learning, Vol. 70 of Proceedings of Machine Learning Research, 2017, pp. 1321–1330.
URL <https://proceedings.mlr.press/v70/guo17a.html>
- [44] P. Koehn, Statistical significance tests for machine translation evaluation, in: Proceedings of the 2004 Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics, Barcelona, Spain, 2004, pp. 388–395.
URL <https://aclanthology.org/W04-3250/>